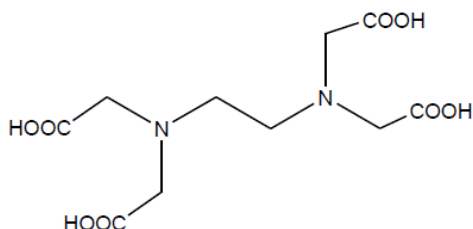


Experiment No-2

Objective: Quantitative estimation of Ca^{+2} and Mg^{+2} ion by complexometric titration using Na_2EDTA

Numerous methods are available for titrimetric determination of various cations with certain organic reagents called complexones. These complexones are imino-polycarboxylic acids, having excellent complex forming ability with a number of cations. The simplest of the complexones is iminodiacetic acid, $\text{HN}(\text{CH}_2\text{COOH})_2$. Other complexones can be assumed to be higher derivatives of this family. The most important member of this family of reagents is ethylenediaminetetraacetic acid, abbreviated as EDTA. The structure of EDTA is



In complexometric determination of magnesium and calcium ions in their mixture, EDTA is used as a titrant and Solochrome Black (Eriochrome Black **T**) as an indicator. EDTA is only slightly soluble in water. However, its disodium salt is freely soluble in water. EDTA, generally, forms **1:1** complexes with metal ions. In reactions, EDTA and its disodium salt are represented as H_4Y and $\text{Na}_2\text{H}_2\text{Y}$, respectively.

Reaction of the disodium salt with a bivalent cation can be written as follows:



It is apparent from the above equation that there is always a competition in solution between the metal ions and the hydrogen ions in seeking the negative sites on EDTA. The equilibrium condition is determined by the strength of the bond between the metal ion and the ligand, and the relative concentrations of metal ions versus hydrogen ions. In other words, we can say that the stability of the metal-EDTA complex will be governed by the hydrogen ion concentration or pH of the solution. Stability of Ca^{+2} and Mg^{+2} complex of EDTA is maximum at $\text{pH} = 10$.

When indicator solution, which is blue in color, is added to the solution containing magnesium and calcium ions, wine red colored metal ion-indicator complexes of varying stability are formed. The magnesium-indicator complex is more stable than the calcium indicator complex but less stable than the

magnesium-EDTA complex which in turn is less stable than the calcium-EDTA complex. Consequently, when EDTA solution is added, it reacts first with the free calcium ions, then with the free magnesium ions, then with the calcium indicator complex and finally with the magnesium-indicator complex. Since the magnesium-indicator complex is wine-red in color and the free indicator is blue between pH 7 and 11, the colour of the solution changes from wine-red to blue at the end point. In this experiment, we titrate one portion of the test solution containing both magnesium and calcium ions with EDTA using Solochrome Black indicator at pH 10. Then another equal portion of test solution is titrated with EDTA under strongly alkaline condition using Patton-Reeder's indicator. In strong alkaline medium Mg is precipitated as $\text{Mg}(\text{OH})_2$ leaving only Ca^{+2} ion in solution. From the amount of EDTA consumed in the 1st titration total amount of Ca^{+2} and Mg^{+2} ions and from the 2nd titration amount of Ca^{+2} ion present in the given unknown solution is obtained. From the difference in the EDTA titre value amount of Mg^{+2} ions will be obtained.

Preparation of solutions:

1. 0.01 M $\text{Zn}(\text{OAc})_2 \cdot 2 \text{H}_2\text{O}$ solution as primary standard: 0.5475 g of A.R. grade $\text{Zn}(\text{OAc})_2 \cdot 2 \text{H}_2\text{O}$ is accurately weighed out in a 250 ml volumetric flask containing 2 g of NH_4Cl dissolved in 10 ml deionised water, the volume is made up to the mark and made uniform by shaking.

2. NH_3 - NH_4Cl buffer (pH = 10) solution:

17.5 g of NH_4Cl is mixed with 142 ml conc. NH_3 (Sp. Gravity 0.88-0.90) and the mixture is made up to 250 ml with deionised water.

3. 0.05 g EBT indicator and 4.9 g KNO_3 mixture is grinded in mortar.

4. Sample solution: It can be prepared by dissolving accurately 200-300 mg of calcium chloride and 100-200 mg of MgSO_4 into minimum quantity of dil. HCl and making up the volume to 250 ml with distilled water.

Standardisation of EDTA solution:

25 ml of the standard $\text{Zn}(\text{OAc})_2 \cdot 2 \text{H}_2\text{O}$ solution is pipetted out into a 250 ml conical flask, 5 ml pH 10 buffer solution is added, diluted to 100 ml with deionised water, warmed to about 40°C ; ~50 mg of EBT indicator are added and the mixture is shaken to obtain a wine-red color. The mixture is then titrated with standard EDTA solution till the color changes to blue. The titration is repeated to get concordant results.

Estimation of total amount of Ca^{+2} and Mg^{+2}

25 ml of the sample solution (acidic in nature) is pipetted out into a 250 ml conical flask, 75 ml de-ionised water is added. Addition of 1-2 drops of methyl red indicator turns the solution red which is neutralised by drop wise addition of 1:1 ammonia until just yellow color develops. Then 5 ml of NH_4Cl - NH_3 (pH= 10) buffer solution and 30-40 mg of EBT indicator are added and the mixture is shaken to obtain a wine-red color. The mixture is then titrated with standard EDTA solution till the color changes to blue.

Estimation of Ca^{+2}

25 ml of the sample solution (acidic in nature) is pipetted out into a 250 ml conical flask, 75 ml de-ionized water is added. Addition of 1-2 drops of methyl red indicator turns the solution red which is neutralized by drop wise addition of 1:1 ammonia until just yellow color develops. 10 ml 10% NaOH solution is added and the mixture is shaken to precipitate $\text{Mg}(\text{OH})_2$ completely. 30-40 mg of Patton-Reeder's indicator is added. The mixture is shaken to obtain wine red color and titrated with standard EDTA until color changes to blue.

Observations:**1. Standardisation of EDTA:**

S.No.	Initial Reading(ml)	Final reading(ml)	Volume of 0.01 M Zinc acetate solution taken (ml)	Volume of E.D.T.A. Solution required (V ml)
1			25	
2			25	
3			25	

2. Determination of total amount of Ca⁺² and Mg⁺²

S.No.	Initial Reading(ml)	Final reading(ml)	Volume of Unknown sample taken (ml)	Volume of E.D.T.A. Solution used up (V ₁ ml)
1			25	
2			25	
3			25	

3. Determination of Ca⁺² only

S.No.	Initial Reading(ml)	Final reading(ml)	Volume of Unknown sample taken (ml)	Volume of E.D.T.A. Solution used up (V ₂ ml)
1			25	
2			25	
3			25	

Calculation:**Standardisation of EDTA:**

$$V_{\text{edta}} \cdot S_{\text{edta}} = V_{\text{ZnOAc}} \cdot S_{\text{ZnOAc}}$$

$$S_{\text{edta}} = (V_{\text{ZnOAc}} \cdot S_{\text{ZnOAc}}) / V_{\text{edta}}$$

$$= f \text{ (M)}$$

Determination of Ca⁺² and Mg⁺² ion in solution

$$1000 \text{ ml of } 0.01 \text{ M EDTA} \equiv 1000 \text{ ml } 0.01 \text{ M Ca}^{+2}$$

$$\equiv 0.40 \text{ g of Ca}^{+2}$$

$$1 \text{ ml of } 0.01 \text{ M EDTA} \equiv 0.0004 \text{ g of Ca}^{+2}$$

$$V_2 \text{ ml f (M) EDTA} = a \text{ g of Ca}^{2+}$$

$$\text{Amount of Ca}^{2+} \text{ in } 25 \text{ ml} = a \text{ g}$$

$$\text{Amount of Ca}^{2+} \text{ in g/L} =$$

Amount of Mg^{2+} in 25 ml = $(v_1 - v_2)$ ml

$$\begin{aligned} 1000 \text{ ml of } 0.01 \text{ M EDTA} &\equiv 1000 \text{ ml } 0.01 \text{ M Mg}^{+2} \\ &\equiv 0.2432 \text{ g of Mg}^{+2} \end{aligned}$$

$(V_1 - V_2)$ ml of (M) EDTA = b g of Mg^{2+}

Amount of Mg^{2+} in 25 ml = b g

Amount of Mg^{2+} in g/L =

Result: